

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

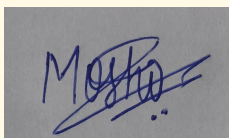
Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	DSA0305	Course Title:	Natural Language Processing
CO Assessed:	CO2 – Manipulation	Assessment:	Assessment Tool 1 – ANALYTICAL PROBLEM SOLVING
Weightage:	50% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
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Section / Batch:	B.Tech (AI-DS)	Date:	01.08.2026

Code of Conduct

I Moshika M (192424064) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the Student:

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
English Word Classes, Part of Speech Tagging	Morphological decomposition of connected, connecting, and connection; identification of roots, suffixes, inflectional vs. derivational forms, and normalization for document indexing.	Q1
English Word Classes, Tagsets for English, Rule-Based Part of Speech Tagging	Morphological parsing of unhappy, happiness, and happily; identification of prefixes, suffixes, base forms, and semantic effects of derivational morphology.	Q2
Counting Words, Part of Speech Tagging, English Word Classes	Stemming and root extraction for played, player, and playing; handling grammatical variations and word formation changes for search preprocessing.	Q3
Finite-State Morphological Analysis, Rule-Based Part of Speech Tagging, Transformation-Based Tagging	Finite-state parsing of writes, writing, and written; modeling regular and irregular verb inflections and root normalization.	Q4
English Word Classes, Rule-Based Part of Speech Tagging, Counting Words	Porter Stemmer-based processing of relational, relation, and relate; application of derivational suffix removal and stem extraction for retrieval.	Q5

Annexure A – ANALYTICAL PROBLEM SOLVING

1. Given the input words: “connected”, “connecting”, “connection”, design a morphological analysis pipeline for a document indexing system.

Apply rules to decompose each word into root and suffix components.

Differentiate between inflectional and derivational forms within the dataset.

Normalize all variants to a common base form for efficient retrieval.

Determine the parsed structure and normalized output for each word.

ANSWER:

Input:

connected, connecting, connection

Rules

- **-ed** → Past tense (Inflectional)
- **-ing** → Present participle (Inflectional)
- **-ion** → Noun formation (Derivational)

Word	Root	Suffix	Type	Normalized Form
connected	connect	-ed	Inflectional	connect
connecting	connect	-ing	Inflectional	connect
connection	connect	-ion	Derivational	connect

Parsed Structure

- connected → connect + ed
- connecting → connect + ing
- connection → connect + ion

Final Output

connected → connect

connecting → connect

connection → connect

2. Given the input words: “unhappy”, “happiness”, “happily”, design a morphological parsing module for sentiment analysis.

Identify prefixes, suffixes, and base forms using rule-based decomposition.

Ensure semantic consistency across derived word forms.

Classify each transformation as inflectional or derivational.

Produce the root and morphological breakdown for each word.

ANSWER:

Input:

unhappy, happiness, happily

Rules

- **un-** → Negative prefix
- **-ness** → Noun suffix
- **-ly** → Adverb suffix

Word	Prefix	Root	Suffix	Type
unhappy	un	happy	—	Derivational
happiness	—	happy	-ness	Derivational
happily	—	happy	-ly	Derivational

Morphological Breakdown

- unhappy → un + happy
- happiness → happy + ness
- happily → happy + ly

Semantic Consistency

All words are derived from the same base:

happy

Final Output

unhappy → happy

happiness → happy

happily → happy

3. Given the input words: “played”, “player”, “playing”, design a stemming-based preprocessing module for a search engine.

Apply morphological rules to strip affixes and identify the base form.

Handle both grammatical variations and word formation changes.

Ensure consistent root extraction across all forms.

Determine the stem and classification for each input word.

ANSWER:

Input:

played, player, playing

Rules

- Remove **-ed**
- Remove **-er**
- Remove **-ing**

Word	Stem	Removed Affix	Type
played	play	-ed	Inflectional
player	play	-er	Derivational
playing	play	-ing	Inflectional

Result

played → play

player → play

playing → play

4. Given the input words: “writes”, “writing”, “written”, design a finite-state morphological parsing module for a text processing system.
Apply state transitions to represent suffix transformations and irregular forms in verb inflection.
Differentiate between regular and irregular inflectional patterns within the dataset.
Ensure consistent mapping of all forms to a common root representation.
Determine the state transitions, morphological breakdown, and normalized output for each word.

ANSWER:

Input:

writes, writing, written

State Diagram (Text Form)

```
Start
|
write
|
|-- +s -----> writes
|
|-- +ing ----> writing
|
|-- irregular ----> written
```

Morphological Breakdown

Word	Breakdown	Pattern	Normalized Form
writes	write + s	Regular Inflection	write
writing	write + ing	Regular Inflection	write
written	write + en	Irregular Inflection	write

State Transitions

- Start → write → +s → writes
- Start → write → +ing → writing
- Start → write → irregular (+en) → written

Final Output

writes → write
writing → write
written → write

5. Given the input words: “relational”, “relation”, “relate”, design a Porter Stemmer-based preprocessing module for document retrieval.
Apply stemming rules step-by-step to remove derivational suffixes and obtain base forms.
Handle variations in suffix patterns while preserving semantic consistency.
Ensure uniform root extraction across all derived forms.
Determine the stemming steps, intermediate forms, and final stem for each input word.

ANSWER:

Input:

relational, relation, relate

Stemming Steps

Word	Step	Intermediate Form	Final Stem
relational	Remove -ational → -ate	relate	relat
relation	Remove -ion	relat	relat
relate	Remove final -e	relat	relat

Explanation

relational

relational
↓ remove -ational
relate
↓ remove -e
• relat

relation

relation
↓ remove -ion
• relat

relate

relate
↓ remove -e
• relat

```
relational → relat
relation  → relat
relate   → relat
```

Numerical Problems					
Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Q. No. / Criteria Level	Problem Understanding	Method Selection	Solution Process	Accuracy of Calculation	Interpretation of Results	Sub. Total	Total %
1							
2							
3							
4							
5							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____